

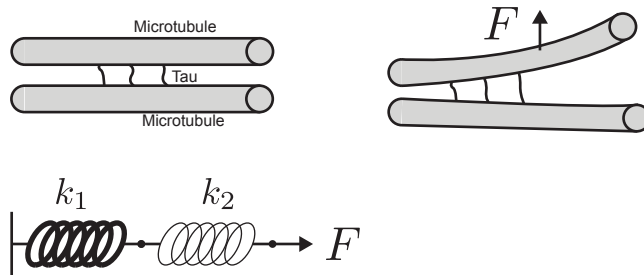
Problem Set 1: What happens in a composite material?

Physics 230A/146A — Biological Physics

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Microtubules are rigid polymers that, among other things, provide mechanical strength inside nerve axons. Several microtubules are bundled together by bundling proteins, one of which is called Tau. In healthy axons, Tau exists as individual filaments that are relatively soft compared to microtubule. In a class of disease called Tauopathies — of which Alzheimer's is the most famous — Tau forms neurofibrillar tangles in which several Tau molecules tangle together. In either case, the mechanical properties of the axon are determined by two objects with different mechanical properties attached in series (stiff microtubules and soft healthy Tau, or stiff microtubules and even stiffer tangled Tau).

In this problem, we ask, if the axon is subject to a force, which mechanical element “takes a bigger hit”?



Suppose a stiff spring with spring constant k_1 is connected, in series, to a soft spring of stiffness $k_2 = \epsilon k_1$ where $\epsilon < 1$. Assume each spring has a linear spring energy $E_i = \frac{1}{2}k_i(\bar{x}_i - x_i)^2$ where $\bar{x}_i$ is the rest length of spring i . Now suppose an extensional force F is applied to both ends of the pair.

- Write the energy of the system $E(x_1, x_2)$. Include the external force F using a fictitious potential. Find the mechanical equilibrium (x_1, x_2) by minimizing the energy. Tip: You could define new configuration variables $\Delta x_i = \bar{x}_i - x_i$ to make algebra cleaner.

- ii. Which spring is stretched more? By how much?
- iii. Which spring experiences a higher tension? By how much?
- iv. Which spring stores more elastic energy? By how much?